Md Shamiul Islam Riyad

AI Engineer (Early Career) · Python · LLM Applications · RAG

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SUMMARY

Early-career AI engineer and CSE undergraduate who builds LLM applications end to end: Python RAG services on Qdrant and Gemini, LLM automations with schema-checked output and human approval, and Bangla and English assistants. Backed by practical software engineering and automated tests; writes step-by-step guides so others can learn from each build.

EDUCATION

BSc in Computer Science and Engineering, United International University (UIU), Dhaka — 2023 – 2027 (expected)

AI coursework: k-nearest neighbours, k-means, A* search and a CSP solver, implemented from scratch in Python.

AI PROJECTS

FactoryBrain – AI operations platform for garment factories — github.com/shamiulriyad/FactoryBrain

Python, FastAPI, LangChain, Qdrant; Gemini, OpenAI or Anthropic (switchable); ASP.NET Core 9, Next.js; 240+ tests. Personal project.

- Connected the backend to a Python RAG service that answers from retrieved manual chunks only, with page citations.
- Built agents that turn live line and machine data into RAG-grounded fixes, with risky results queued for human approval.
- Automated purchase-order and tech-pack extraction with schema-constrained LLM output whose quotes are checked against the source document; added photo-based defect detection with a vision-language model.

ragkit-dev – open-source Python RAG toolkit — github.com/shamiulriyad/rag-learning

Python, LangChain, Qdrant, Hugging Face (BGE-M3), Gemini, FastAPI, Streamlit, pytest, GitHub Actions. Personal project.

- Split ingestion and querying into 11 single-purpose modules so each step can be read, tested and swapped.
- Gemini answers from retrieved context only, citing chunk and PDF page per claim; local (BGE-M3) or Gemini embeddings.
- Pip-installable CLI with a FastAPI service, Streamlit UI and npm project generator; 290+ tests run offline in CI.

rag-kit – multi-tenant RAG platform with chat and citations — github.com/shamiulriyad/rag-kit

Python, FastAPI, Qdrant, Gemini, ASP.NET Core, React, TypeScript, PostgreSQL, Docker Compose. Personal project.

- Built a .NET gateway (auth, plan limits) in front of a Python RAG service, with one Qdrant collection per knowledge base.
- Wrote 12 tutorial-style guides (what, why, how, where in code, how to change, example) on each part of the pipeline.

QueueStorm Investigator – LLM support copilot for digital finance — github.com/shamiulriyad/sust_preli

ASP.NET Core 8, Claude API, Docker. bKash SUST CSE Carnival 2026 hackathon.

- Checks Bangla, Banglish or English complaints against transaction history; returns one JSON verdict with a draft reply.
- Falls back to rules if Claude fails; every reply passes schema validation and a filter that blocks PIN/OTP requests.

Chokh – AI narration for visually impaired users, in Bengali — Live demo

React, ASP.NET Core, Gemini vision, Web Speech API, Docker. ML Bangladesh Hackathon 2026.

- Deployed app that sends camera frames to Gemini every ~2.5 s and speaks hazards first, then position and printed text.
- Wrote the Bengali system prompt to admit uncertainty instead of inventing objects and to refuse medical advice.

OTHER PROJECTS

- **NogorSathi AI:** team Gemma 4 commuter-agent prototype; built the demo that traces each tool call and route choice.
- **Doctor MCP Server:** list and call endpoints exposing 5 scheduling tools to an LLM assistant, over MySQL with Zod validation.
- **AI Study Assistant:** ask by text or voice, get a simple Bangla explanation from Gemini, read aloud in the browser.
- **ShilpoHubBD:** team of 5; wrote 790 of 964 backend commits (ASP.NET Core, PostgreSQL), merged 96 pull requests.

TECHNICAL SKILLS

AI and LLMs: Gemini, Claude and OpenAI APIs, prompt design, structured output, RAG, embeddings, LangChain, Qdrant
Languages: Python, C#, TypeScript, JavaScript, C++, SQL
Frameworks and tools: FastAPI, Streamlit, ASP.NET Core, React, Next.js, PostgreSQL, Docker
Testing and CI: pytest, xUnit, Playwright, GitHub Actions, Git
Currently learning: PyTorch and computer vision fundamentals

ACHIEVEMENTS

- Featured in Google Developer Experts AI Community Highlights, Build with Gemma @ Bangladesh (Jul 2026).
- Top 70 of 700+ teams, Infinity AI BuildFest (Jun 2026); 2nd runner-up, UIU CSE Project Show (Mar 2026).
- 7 hackathons in 2026, including the IUT ICT Fest Bdapps Agentic AI Hackathon and the bKash SUST CSE Carnival.
- 575+ problems solved in 55+ contests (LeetCode 1562, CodeChef 1448); 4th place, UIU Team Contest (Jul 2026).